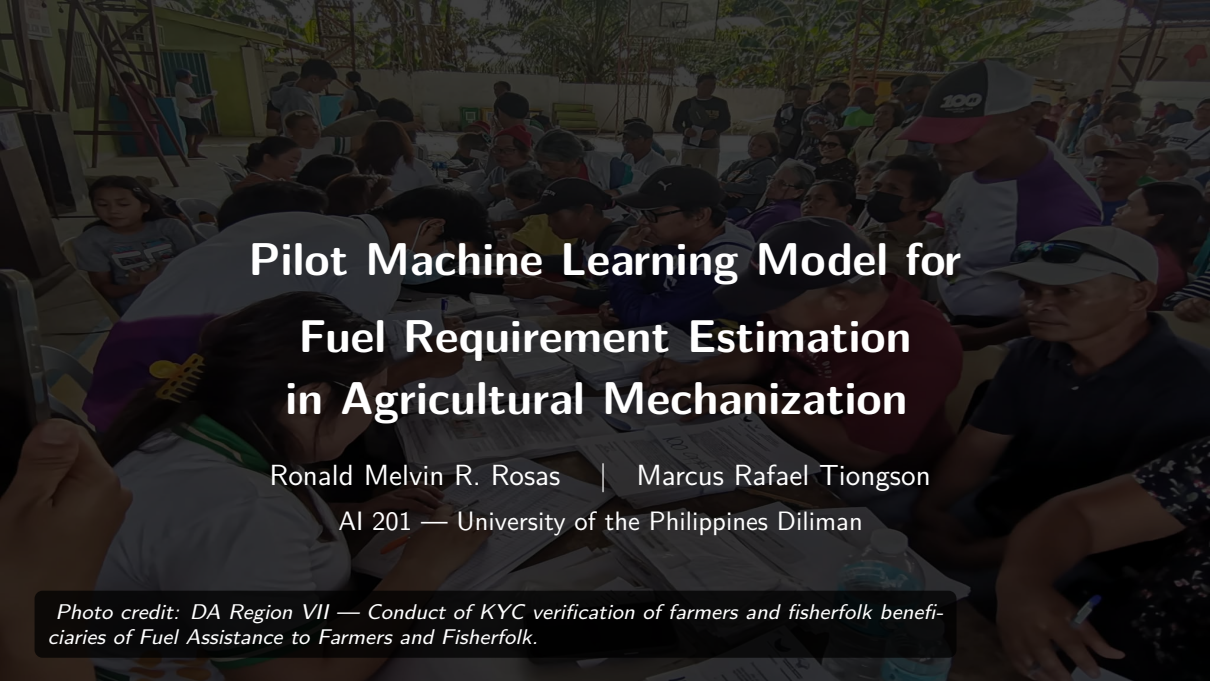




Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

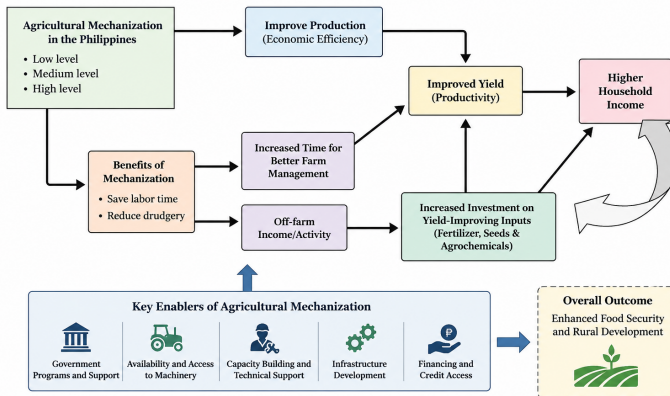
Ronald Melvin R. Rosas | Marcus Rafael Tiongson

AI 201 — University of the Philippines Diliman

Photo credit: DA Region VII — Conduct of KYC verification of farmers and fisherfolk beneficiaries of Fuel Assistance to Farmers and Fisherfolk.

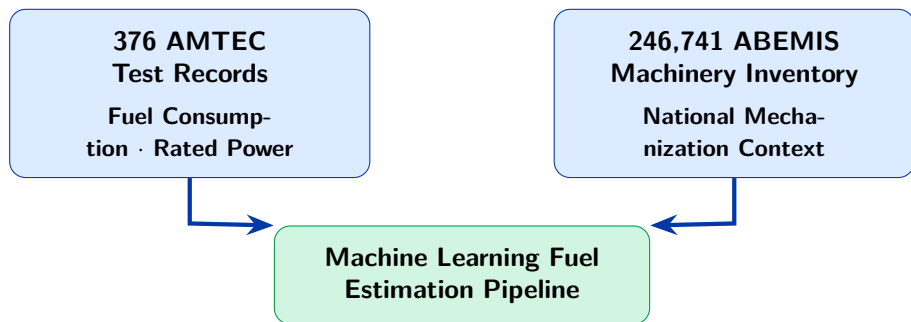
Agricultural Mechanization in the Philippine Context

Philippine Agricultural Mechanization Context



Fuel-dependent agricultural operations require accurate and scalable fuel requirement estimation methods for mechanization planning, government support programs, and inventory-based decision support.

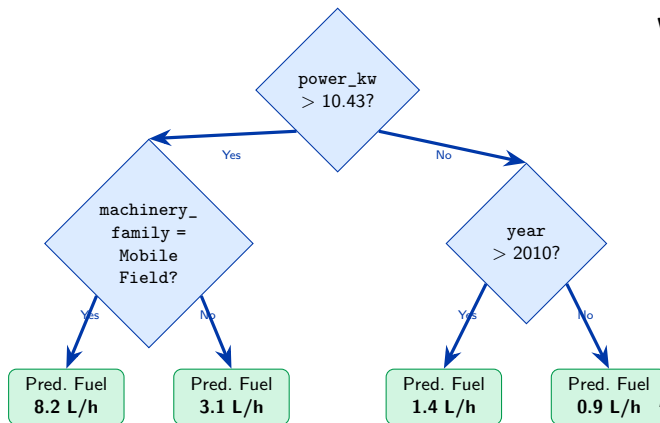
The Data: AMTEC & ABEMIS



AMTEC — Philippine machinery testing and evaluation center, based in UPLB, providing standardized agricultural machinery performance and fuel consumption data.
Source: UPLB-AMTEC (1990s–2026)

ABEMIS — National agricultural engineering information system of BAFE for machinery inventory, infrastructure monitoring, and mechanization decision support.
Coverage: all 18 Philippine regions

The Engine: Capturing Non-Linear Physical Interactions



Why not OLS?

- Power and fuel do *not* scale linearly across machine types
- Each tree partitions the feature space differently
- 300 trees average out variance — robust predictions
- No manual feature crossing required

$$R^2_{\text{RF}} = 0.916 \quad \text{vs.} \quad R^2_{\text{OLS}} = 0.704$$

Random Forest vs. OLS Baseline ($n_{\text{test}} = 76$)

Model	R^2	MAE (L/h)	RMSE (L/h)
OLS Global Linear	0.704	1.882	2.875
OLS Hierarchical Fallback	0.703	1.879	2.879
Random Forest	0.916	0.950	1.534

+30%

better R^2

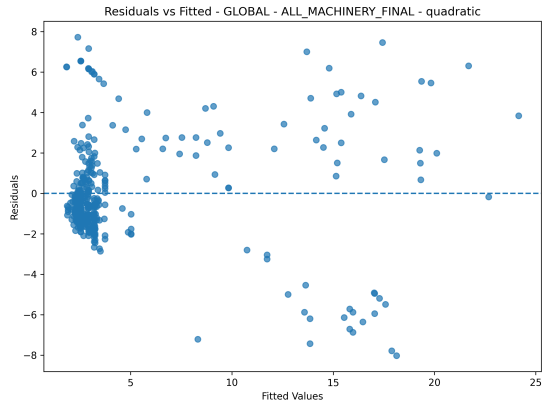
2×

lower MAE

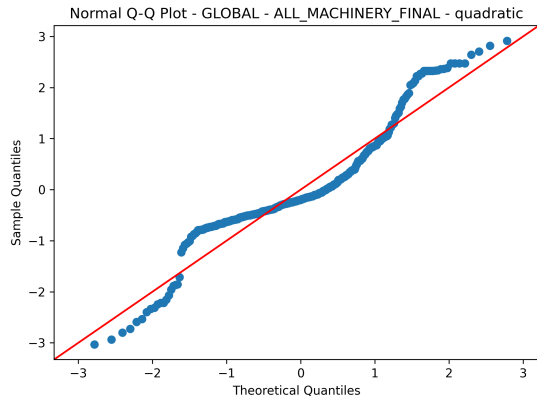
2×

lower RMSE

OLS Diagnostics — Global (All Machinery)



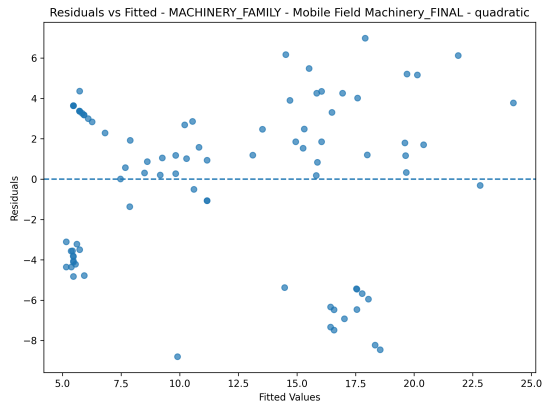
Residuals vs. fitted



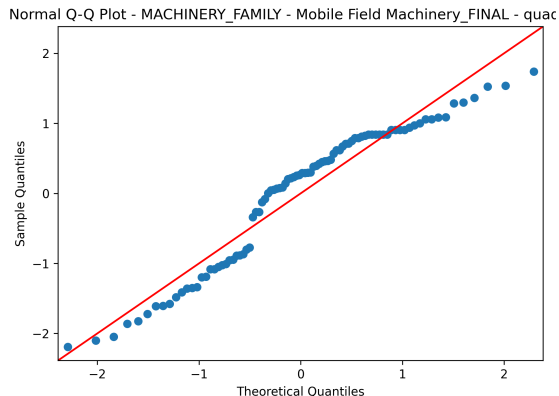
Q-Q plot

Residuals fan out at the high-power tail and the Q-Q deviates in the upper quantiles — the global quadratic OLS underfits heavy mobile machinery, motivating the per-family fallback and the move to Random Forest.

OLS Diagnostics — Mobile Field Machinery



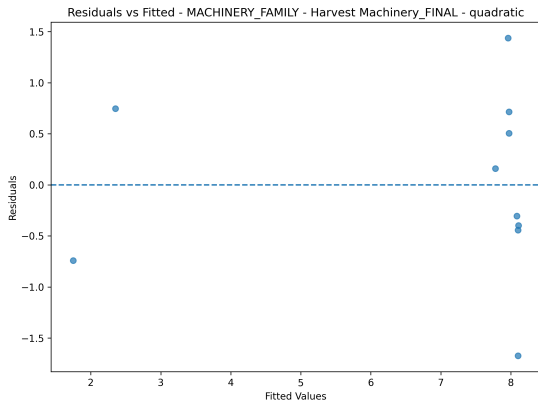
Residuals vs. fitted



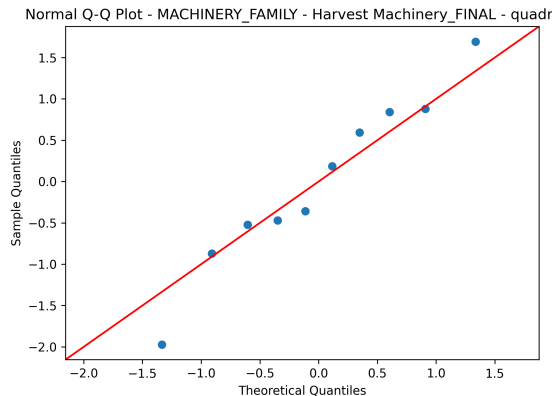
Q-Q plot

Within the Mobile Field family (tractors, tillers) the quadratic fit is markedly tighter — residuals are more symmetric and the Q-Q tracks normal across most of the range, validating the per-family scope as the most useful OLS slice.

OLS Diagnostics — Harvest Machinery



Residuals vs. fitted



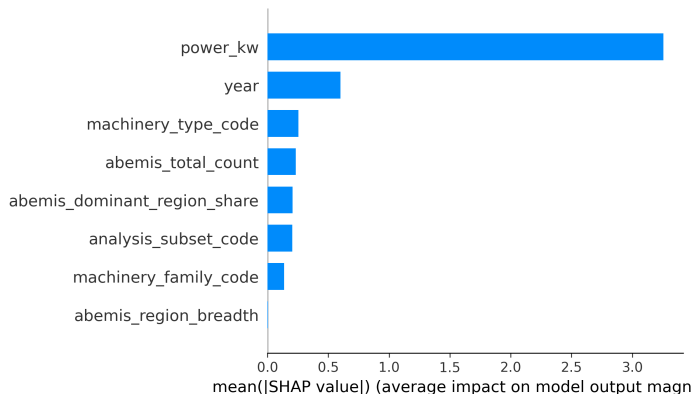
Q-Q plot

Harvest machinery (combines, reapers) holds a serviceable quadratic shape but with a heavier upper-quantile Q-Q tail than Mobile Field — a residual non-linearity that Random Forest captures without needing manual feature engineering.

But why does the model trust these predictions?

Opening the black box with SHAP and LIME.

What Drives the Model? (Validating Physics with SHAP)



Key finding:

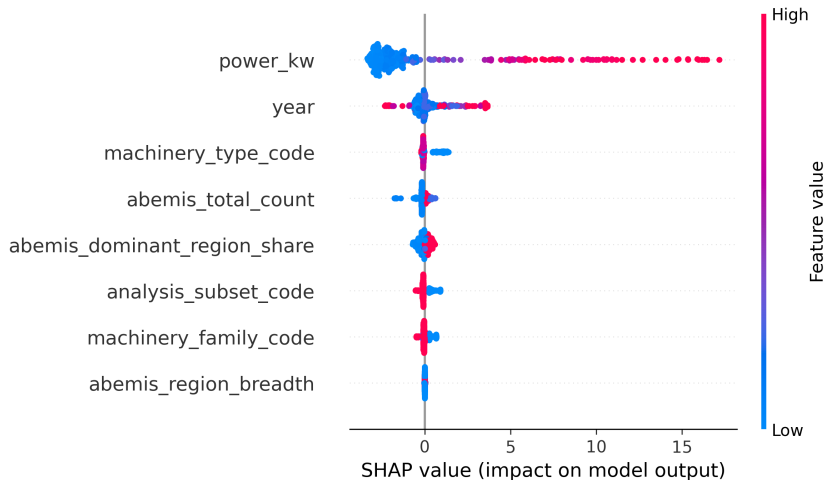
power_kw accounts for **~66.5%** of total prediction importance.

year (equipment vintage) is second at **~12.3%**, reflecting improved combustion efficiency in newer equipment.

ABEMIS context features contribute a combined **~9%**, adding real but secondary value.

The model didn't memorize data — it *learned physics*.

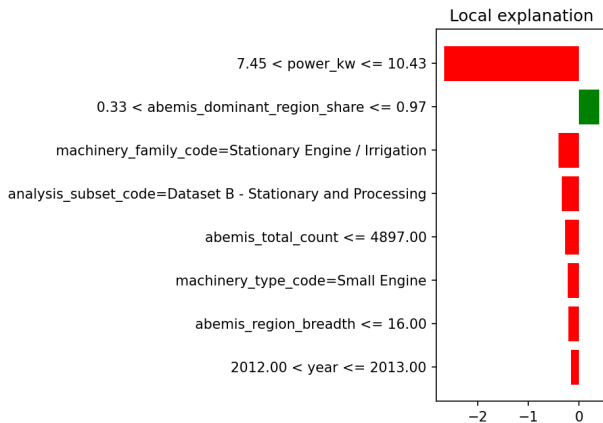
SHAP Beeswarm — Distribution of Feature Contributions



Each dot is one test record; color encodes feature value (red = high, blue = low). High `power_kw` consistently pushes predictions up; `year` contributions are bidirectional, capturing both older high-fuel and modern efficient cohorts.

Explaining Individual Predictions (LIME)

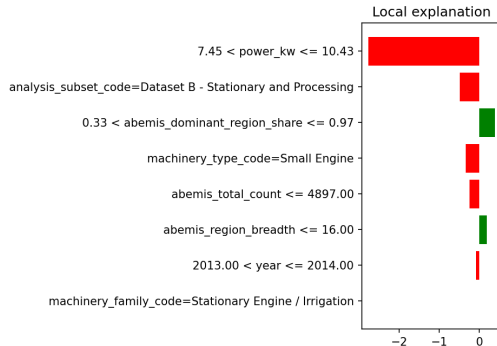
LIME - Sample 372 | Small Engine | Actual=5.240 Pred=4.773



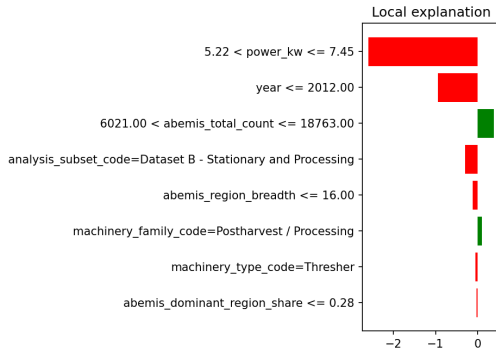
LIME breaks down *why* the model predicted **4.77 L/h** for one specific small engine — granular transparency that builds trust for BAFE engineers.

LIME — More Local Explanations Across Machinery Types

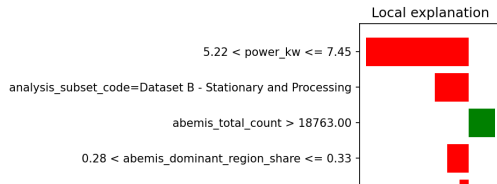
LIME - Sample 284 | Small Engine | Actual=3.560 Pred=3.670



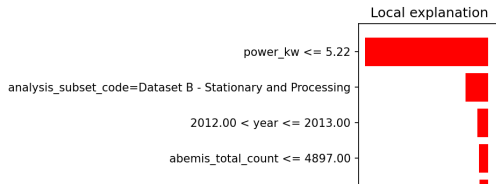
LIME - Sample 32 | Thresher | Actual=1.570 Pred=1.601



LIME - Sample 241 | Water Pump | Actual=2.330 Pred=2.115



LIME - Sample 162 | Small Engine | Actual=1.770 Pred=1.972



From Lab to National Scale

172,265

machines successfully scored

National inventory: 246,741

Coverage: **70%**

Missing power data: 62,808

Unmappable type: 11,668

Outputs per region:

- Per-record fuel estimate (L/h)
- Per-region aggregate demand
- Per-region $\times$ machinery-type breakdown

The unscored 30% reflects *data gaps* — not model failure. Closing those gaps is part of the roadmap.

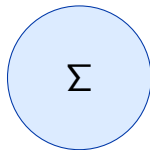
An aerial photograph of a village nestled in a valley, surrounded by lush green mountains. The village features a cluster of buildings with colorful roofs (red, blue, green). Beyond the village, the landscape is dominated by extensive terraced rice fields that follow the contours of the hillsides. The background shows steep, forested mountains under a clear sky.

What's next?

Closing the data gaps. Validating in the field.

Future Directions of the Study

The proposed framework may be expanded toward national-scale fuel forecasting, seasonal planning, and data-driven mechanization support.



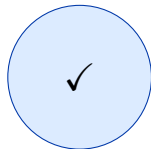
National Fuel Forecasting

Estimate regional and seasonal agricultural fuel demand using national machinery inventories and operational datasets.



Seasonal Planning

Integrate cropping calendars to identify peak planting and harvesting fuel demand periods.



Real-World Validation

Compare machine learning predictions against actual farm-level fuel consumption observations.

Long-term applications may support fuel assistance programs, mechanization planning, and agricultural decision support systems in the Philippines.

Thank you

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